

Evaluating a Postfix Expression:

Each operator in a postfix string refers to the previous two operands in the string.

Suppose that each time we read an operand we push it onto a stack. When we read an operator, its operands will be the top two elements; perform the indicated operation on them and push the result on the stack so that it will be available as an operand of the next operator.

The algorithm for postfix evaluation is given below:

S = the empty stack;

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while (the input string is not reached end)
{
    symbol = next input character;

    if (symbol is an operand)
        push (S, symbol);

    else /* symbol is an operator */
    {
        operand2 = pop(S); /* pop 2 values */
        operand1 = pop(S);
        result = operand1 (operator) operand2;
        push(S, result);
    }
}

return(pop(S));
```

At the end, the remaining value on the stack is the result.